

Foundation (Jalapeno)

Open Ended Questions (FRQ)

- Q9.** A space probe is traveling through the solar system. Its distance from the sun is given by the equation $y = 3x - 2$, where y is the distance in millions of kilometers and x is the time in months. If the probe is currently 10 million kilometers away from the sun, how many months will it take for the probe to be 25 million kilometers away? Show your work and explain your answer.
- Q10.** A lab experiment involves mixing two substances, A and B . The amount of substance A required is given by the equation $y = 2x + 1$, where y is the amount in grams and x is the temperature in degrees Celsius. The amount of substance B required is given by the equation $y = x - 3$. If the temperature is currently 10 degrees Celsius, and 5 grams of substance A are required, how many grams of substance B are required? Show your work and explain your answer.

Chef's Tips

- Tip 1: Remind students to read the questions carefully and identify the variables and constants.
- Tip 2: Encourage students to use visual aids such as graphs and tables to help them solve the problems.
- Tip 3: Emphasize the importance of checking the reasonableness of their answers.